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PATENTS ACT, 1978

CERTIFICATE

In accordance with section 44 (1) of the Patents Act, No. 57 of 1978, it is hereby certified that:

DR. GIRENDRA KUMAR GAUTAM; DR. SHIVENDRA AGARWAL; DR. DIMAK CHAND SAHU; DR. SATENDRA KUMAR; DR. VIRENDRA KUMAR PATEL; DR. DEEPAK SHARMA; DR. JOOHEE PRADHAN; DR. SUNITA PANCHAWAT; MS. DEVSHREE GAYAKWAD; DR. SWETA SHRIVASTAVA KOKA; MS. ANAMIKA SINGH; MS. BHAGYASHREE AGARWAL

Has been granted a patent in respect of an invention described and claimed in complete specification deposited at the Patent Office under the number

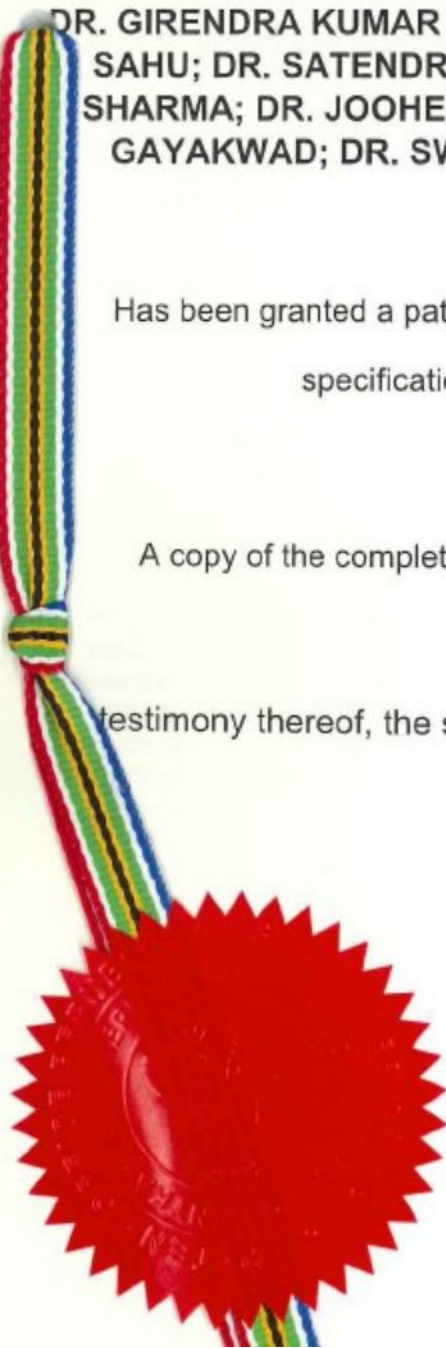
2022/03599

A copy of the complete specification is annexed, together with the relevant Form P2.

In testimony thereof, the seal of the Patent Office has been affixed at Pretoria with effect from the 31st day of August 2022

A handwritten signature in black ink, appearing to read 'J. J. J.', written over a circular stamp.

Registrar of Patents



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PATENTS ACT, 1978
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FORM P2

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Priority claimed:		Country	Number	Date	
54 Title of invention					
ATORVASTATIN ETHOSOMES TOPICAL GEL BASED DRUG DELIVERY SYSTEM					
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REPUBLIC OF SOUTH AFRICA
PATENTS ACT, 1978
COMPLETE SPECIFICATION
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FORM P7

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TITLE OF INVENTION

54	ATORVASTATIN ETHOSOMES TOPICAL GEL BASED DRUG DELIVERY SYSTEM
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ATORVASTATIN ETHOSOMES TOPICAL GEL BASED DRUG DELIVERY SYSTEM

FIELD OF INVENTION

- 5 The present invention relates to develop a ethosomal formulation of atorvastatin to reduce the first pass metabolism and to enhance its systemic bioavailability.

BACKGROUND OF THE INVENTION

- 10 “Ethosomes are ethanolic liposomes”. Ethosomes can be defined as noninvasive delivery carriers that enable drugs to reach deep into the skin layers and/or the systemic circulation. Ethosomes are utilized mainly for the enhanced delivery of medicinal agent across the skin membrane. In return to the need of advanced drug delivery system Touitou invented a vesicular system called ethosomes for topical drug delivery across the skin. Ethosomes are soft, smoothy, malleable vesicles utilized for novel delivery of medicaments.

- 15 Atorvastatin (Lipitor) is a member of the drug class known as statins, it is a HMG-CoA reductase inhibitor. It is used for lowering cholesterol. Atorvastatin is a competitive inhibitor of hydroxymethylglutaryl-coenzyme A (HMG-CoA) reductase, the rate-determining enzyme in cholesterol biosynthesis via the mevalonate pathway. Atorvastatin acts primarily in the liver. The absolute bioavailability of atorvastatin is approximately 14% and the systemic bioavailability of HMG-CoA reductase inhibitory activity is approximately 30%. The low systemic bioavailability is due to presystemic clearance by gastrointestinal mucosa and first pass metabolism in the liver. Atorvastatin is most widely used in coronary heart disease (CHD).

- 25 The I.P. defines Gels are homogeneous, semisolid preparations usually consisting of solutions or dispersions of one or more medicaments in suitable hydrophilic or hydrophobic bases. They are normally prepared with the aid of suitable gelling agents like HPMC, Carbopol and Sodium CMC etc. Substances such as antioxidants, stabilizers and antimicrobial preservatives are used as additives in the formulation of gels.

OBJECTS OF THE INVENTION

5 Some of the objects of the present disclosure, which at least one embodiment herein satisfies, are as follows.

It is an object of the present disclosure to ameliorate one or more problems of the prior art or to at least provide a useful alternative

10 An object of the present disclosure is to provide a to develop a ethosomal formulation of atorvastatin to reduce the first pass metabolism and to enhance its systemic bioavailability.

15 Still another object of the present disclosure is to provide that ethosomal formulation was designed to achieve high permeability.

Another object of the present disclosure is to provide that atorvastatin was formulated into ethosomes and then formulated into gel.

20 Another object of the present disclosure is to provide that the gel showed enhanced bioavailability as compared to normal plane drug gel.

Still another object of the present disclosure is to provide that by changing the ratio of Phospholipid and ethanol, atorvastatin release can be controlled for a prolonged period of time by reducing possible side effects.

25 Other objects and advantages of the present disclosure will be more apparent from the following description, which is not intended to limit the scope of the present disclosure.

SUMMARY OF THE INVENTION

30 The following presents a simplified summary of the invention in order to provide a

basic understanding of some aspects of the invention. This summary is not an extensive overview of the present invention. It is not intended to identify the key/critical elements of the invention or to delineate the scope of the invention. Its sole purpose is to present some concept of the invention in a simplified form as a prelude to a more detailed description of the invention presented later.

The present invention is generally directed to develop a ethosomal formulation of atorvastatin to reduce the first pass metabolism and to enhance its systemic bioavailability.

An embodiment of the present invention is atorvastatin was formulated into ethosomes and then formulated into gel.

An embodiment of the present invention is atorvastatin was initially formulated into ethosomes using soya lecithin a phospholipid and ethanol as permeation enhancer and then finally formulated into gel utilizing carbopol 934 as gelling agent.

Another embodiment of the invention is Initially formulated atorvastatin ethosomes were evaluated for particle size, shape, zeta potential, entrapment efficiency and drug release profile.

Yet another embodiment of the invention is the zeta potential values showed that ethosomes were having suitable negative charge to remain as smaller vesicles which makes them compatible to formulate into the gel.

Yet another embodiment of the invention is the so formed ethosomal gel showed enhanced bioavailability as compared to normal plane drug gel and also oral marketed formulation.

Yet another embodiment of the invention is by changing the ratio of Phospholipid and ethanol, atorvastatin release can be controlled for a prolonged period of time.

BRIEF DESCRIPTION OF THE DRAWINGS

Figure 1: Scanning electron microscopy of Atorvastatin Ethosomal preparation.

Figure 2: Zeta potential of prepared Atorvastatin ethosomes.

5 Figure 3: Dissolution profile of Atorvastatin ethosomal gel formulation.

Figure 4: Zero order plot for drug release from ethosomal gel.

Figure 5: First order for drug release from ethosomal gel.

Figure 6: Higuchi model for drug release from ethosomal gel.

DETAILED DESCRIPTION OF THE INVENTION

10 The following description is of exemplary embodiments only and is not intended to limit the scope, applicability or configuration of the invention in any way. Rather, the following description provides a convenient illustration for implementing exemplary embodiments of the invention. Various changes to the described embodiments may be made in the function and arrangement of the elements
15 described without departing from the scope of the invention.

EXAMPLE 1: Preparation of Ethosomes

Atorvastatin loaded ethosomes were prepared by cold method followed by ultra
sonication. Phospholipids, drug and other lipid material were dissolved in ethanol in a
20 covered beaker at a room temperature by vigorous stirring with the use of magnetic
stirrer. Propylene glycol was added during stirring. This mixture was heated to 30⁰C
in a water bath. The water heated to 30⁰C in a separate beaker was added to the mixture
which was then stirred for 5 min in a covered vessel using magnetic stirrer. The vesicle
size of ethosomal formulation was then reduced by ultrasonic probe sonicator for 10
25 min and off time was 10 sec. Finally the formulation was stored under refrigeration.
The vesicles were observed under Carl Zeiss trinocular microscope at 40 X and 100 X
magnification.

Table 1: Formulation of Atorvastatin Ethosomes

Ingredient	Formulation code							
	E1	E2	E3	E4	E5	E6	E7	E8
Atorvastatin (gm)	20	20	20	20	20	20	20	20
Soya Lecithin (ml)	15	15	15	30	30	30	45	60
Cholesterol (ml)	15	30	45	15	30	45	15	45
Ethanol (ml)	6	6	6	9	9	12	12	15
Propylene glycol (ml)	3	3	3	3	3	3	3	3
Distilled water	Upto 30ml	Upto 30ml	Upto 30ml	Upto 30ml	Upto 30ml	Upto 30ml	Upto 30ml	Upto 30ml

EXAMPLE 2: Incorporation Into Gel

Carbopol 934 forms very good consistency Transparent gel at low concentration.

5 Carbopol 1% w/v was soaked in minimum amount of water for an hour. 20 ml of ethosomal suspensions containing Atorvastatin (100mg) was added to the swollen polymer under the continuous stirring at temperature of 30°C until homogeneous gels were achieved. Triethanolamine was utilized to maintain the pH of the preparation and it was stirred slowly until a clear transparent gel was obtained. Finally the ethosomal

10 gel was mixed using a mechanical stirrer for 5 minutes.

EXAMPLE 3: Preparation of plain drug in gel

Plain gel of Atorvastatin was prepared by similar method applied for preparation of ethosomal gel. In this method, required quantity of Atorvastatin was taken and

triturerated into water miscible gel base. Final concentration of Atorvastatin in formulations was fixed to 0.4%.

Initially formulated atorvastatin ethosomes were evaluated for particle size, shape, zeta potential, entrapment efficiency and drug release profile.

5

EXAMPLE 4: Evaluation of atorvastatin ethosomes

The ethosomes were evaluated for particle size, particle shape, zeta potential and elasticity measurement. Table 2 and fig 1 shows the particle size of various formulation determined by scanning electron microscopy.

10

Table 2: Particle sizes of Atorvastatin ethosomal formulations.

Formulation Code	Particle size
E1	94.64
E2	112.52
E3	153.63
E4	149.75
E5	95.6
E6	117.85
E7	94.62
E8	104.69

Elasticity measurement

The elasticity of ethosomes vesicles were measured by extrusion method. The ethosomal formulation were extruded through filter membrane (pore size diameter- 100 nm), using a stainless steel filter holder having 50 mm diameter, by applying a pressure of 2.5 bar. The quantity of vesicle suspension, extruded in 5 minutes was measured.

15

Table 3: Elasticity of Atorvastatin ethosomal formulations

Formulation Code	Elasticity
E1	84.72

E2	80.93
E3	86.07
E4	86.4
E5	89.9
E6	84.68
E7	84.33
E8	83.91

Vesicle size and zeta potential: Vesicle size of the ethosomes can be determined by Malvern instrument. Zeta potential of the formulation can be measured by Malvern zeta sizer. Zeta potential values of prepared Atorvastatin ethosomes were shown in figure 2. zeta potential values showed that ethosomes were having suitable negative charge to remain as smaller vesicles which makes them compatible to formulate into the gel.

EXAMPLE 5: Drug entrapment Efficiency

The entrapment efficiency of ethosomes can be measured by the ultracentrifugation technique. The entrapment efficiency (EE), of Atorvastatin ethosomes, was determined by measuring the concentration of free Atorvastatin in the dispersion medium. 1.0 ml of the ethosomes was diluted up to 10 ml with phosphate buffer (pH 5.5) and centrifuged at 14000 rpm for 1 hour to separate the lipid and aqueous phase. Supernatant was then filtered by 0.2μ membrane filter and analyzed by UV-VIS spectroscopy at 244 nm.

$$\% \text{ EE} = \frac{[\text{Initial drug} - \text{Free drug}] \times 100}{\text{Initial drug}}$$

Where,

Initial drug = mass of initial drug used for the assay.

Free drug = mass of free drug detected in the supernatant after centrifugation of the aqueous dispersion.

Table 4 indicates the entrapment efficiency of atorvastatin ethosomal formulation. The data indicate that entrapment efficiency depends on ethanol concentration, as the concentration increases up to 30%, results increases in entrapment efficiency of ethosomal formulation.

Table 4: Entrapment efficiency of atorvastatin ethosomal formulations.

Formulation	EE (%)
E1	97.8
E2	96.44
E3	98.58
E4	97.48
E5	98.65
E6	96.45
E7	98.04
E8	96.65

EXAMPLE 6: Drug content

Drug content of the ethosomes can be determined using UV spectrophotometer. This can also be quantified by a modified high performance liquid chromatographic method. Ethosomal suspension (1 ml) was taken and dissolved in 100ml methanol and shaken vigorously for 2 hr in magnetic stirrer, there after the resultant solution was filtered and 1ml of sample was withdrawn and diluted to 10 ml in a volumetric flask and the absorbance was recorded in UV spectrophotometer at 244 nm.

EXAMPLE 7: In-vitro release studies

In-vitro drug release from the formulation was studied by the diffusion cell. Phosphate buffer pH (5.5) is used as diffusion medium, stirred at 50 rpm at 37°C ± 0.5°C. The 5 ml of ethosomal formulations were filled in test tubes and the end of the test tubes was covered with a dialysis membrane and was placed in such way

that it touches the surface of diffusion medium present in receptor compartment. The drug sample were withdrawn at the interval of half an hour for the time period of 12 hrs from diffusion medium and analyzed by using UV spectrophotometer at 244 nm, using phosphate buffer pH (5.5) as a blank.

5 **Table 5: In vitro drug release profile of ethosomal gel formulation and plane drug gel formulation.**

Time (hrs)	% Cumulative drug release	
	Ethosomal gel formulation	Plane drug gel formulation
0	0	0
1	45.53	6.46
2	48.98	8.45
4	55.13	12.56
6	60.97	18.54
8	67.62	25.62
10	73.41	30.43
12	76.26	47.32
24	84.97	59.93
48	98.47	65.37

EXAMPLE 8: Evaluation of Ethosomal Gel

10 **Measurement of pH of the ethosomal gel:** 1 g Atorvastatin ethosomal gel was mixed in 100 ml distilled water with homogenizer. Then the electrode was immersed in the prepared gel solution and readings were recorded from digital pH meter in triplicate and average value was calculated. The pH of gel base and freshly prepared E8 ethosomal gel was found to be 7.3 and 7.9 respectively.

15 **Viscosity study:** Viscosity estimations were performed on Brookfield viscometer by choosing a desirable spindle number and rpm. 50 g of preparation was kept in 50 ml beaker which was set till spindle groove was dipped and rpm was set and dial reading was measured after three minutes. From the reading obtained, viscosity was calculated

by using factor. The procedure was repeated three times and observations are recorded as mean. The viscosity of carbopol 934 gel base and ethosomal gel by brookfield viscometer was found to be 734,00 and 75,100 cps (centipoise).

5 **Spreadability:** It is defined as the extend of area covered when gel is applied over the skin or affected part, as the spreading value effects the therapeutic efficacy of a formulation. Time taken by two slides to slip off from gel in seconds when they placed between the slides under the direction of certain load. Minimum time taken by the slides for separation, shows better spreadability.

10 The spread ability of ethosomal gel was found to be 15.69 g.cm². The spreadability results showed that ethosomal gel was most effective i.e. it showed best result for spread ability.

15 **Extrudability study:** The extrudability of ethosomal gel was determined by filling ethosomal gel in the collapsible tubes. The extrudability of the formulated ethosomal gel was established in terms of weight in grams required to extrude a 0.5 cm ribbon of gel in 10 second. The extrudability of ethosomal gel was found to be positive.

Percentage yield: The empty container was weighed in which the ethosomal gel formulation was stored then again the container was weighed with ethosomal gel formulation. The value of practical yield was expressed as empty container weight was subtracted with the container with gel formulation.

20 **Homogeneity and grittiness:** A minimum quantity of ethosomal gel was pressed between the index finger and thumb. The existance of coarse particles was checked between the figures to notify the consistency of gel. Also, the homogeneity could be detected when a small quantity of the ethosomal gel was rubbed on the skin of the back of the hand. The grittiness of prepared ethosomal gel was also observed in
25 the same manner. Ethosomal gel was found to be homogeneous and no grittiness was noted.

EXAMPLE 9: In vitro release studies

Skin permeation studies Franz diffusion cell was used for permeation studies.

Study was conducted using prepared rat skin. 28 ml of PBS 7.4 was taken in receptor compartment and was continuously stirred with a magnetic stirrer and equilibrated at $37^{\circ}\pm 1^{\circ}\text{C}$ with a recirculating water bath. The prepared skin was assembled with stratum corneum facing upward into the donor apartment. 1 g of ethosomal gel formulation was taken in donor compartment and covered with parafilm to avoid any evaporation process. 5 ml sample was withdrawn through the sampling port at predetermined intervals over 8 hours and each sample was replaced with equal volume of fresh dissolution medium. Then the samples were analyzed for drug content by using phosphate buffer as blank with UV-Visible double beam spectrophotometer at 246 nm. Similar study was performed with marketed atorvastatin gel. Dissolution profile of Atorvastatin ethosomal gel formulation were shown in figure 3.

Table 6: Dissolution data of Atorvastatin ethosomal gel for in vitro release kinetics

Time(hrs.)	%cumulative Release	Log% cumul.Release	% cumul. Drug remaining	Log cumul. % drug remaining
1	45.53	1.65	54.47	1.73
2	48.98	1.69	51.02	1.70
4	55.13	1.74	44.87	1.65
6	60.97	1.78	39.03	1.59
8	67.62	1.83	32.38	1.51
10	73.41	1.86	26.59	1.42
12	76.26	1.88	23.74	1.37
24	84.97	1.92	15.03	1.17
48	98.13	1.99	1.87	0.27

EXAMPLE 10: In Vitro Drug Release Kinetics

In order to analyse the mechanism of release of drug, the release data were analyzed

with the following three mathematical models:

(1) zero-order kinetic

(2) first-order kinetic

(3) Higuchi kinetic

5 **Zero-order model**

According to zero order model drug dissolution from dosage forms do not disaggregate and releases the drug slowly. It is represented by the following equation:

$$Q_t = Q_0 + K_0 t$$

10 In the above equation

- ✓ Q_t = the amount of drug which dissolves in time t ,
- ✓ Q_0 = the initial amount of drug into the solution
- ✓ K_0 = the zero order release constant

15 For zero order model the data obtained from in vitro release study were plotted as a cumulative amount of drug released versus time.

First order model

This model has been used to explain absorption and elimination of some drugs, in theoretical way it is difficult to have concepts for this mechanism. The following equation is mentioned for release of the drug for first order kinetics:

20 $\text{Log } Q_t = \text{Log } Q_0 - \frac{K_1 t}{2.303}$

$$2.303$$

Where ,

- ✓ Q_0 = the initial concentration of drug,
- ✓ K_1 = rate constant for the first order,

✓ t = the time.

The plots were made by log cumulative percentage of drug remaining versus time which gives a straight line with a slope of $-K/2.303$.

Higuchi model

5 This model is based on the hypothesis that

- initial drug concentration is more high than drug solubility;
- diffusion of drug takes place in one dimension only ;
- particles size of drug is much smaller than thickness of system;
- drug diffusivity is constant; and
- 10 ➤ in the release environment a perfect sink condition is always attained.

The higuchi model equation is:

$$Q = K_H t^{1/2}$$

Where,

K_H = the Higuchi dissolution constant,

15 $t^{1/2}$ = square root of time.

The correlation coefficient of release profiles of these models for Atorvastatin ethosomal gel formulation is shown in Table 7. According to the obtained data, first order release profile was found to be the best fit model for our optimized AEGF formulation. The graphs of all three release models for Atorvastatin ethosomal gel
20 formulation are shown in Figures 4, 5 and 6.

Table 7: Drug release kinetic models and their R values for Atorvastatin ethosomal gel

Model	R value
-------	---------

Zero order	0.8224
First order	0.986
Higuchi	0.9546

5 While considerable emphasis has been placed herein on the specific features of the preferred embodiment, it will be appreciated that many additional features can be added and that many changes can be made in the preferred embodiment without departing from the principles of the disclosure. These and other changes in the preferred embodiment of the disclosure will be apparent to those skilled in the art from the disclosure herein, whereby it is to be distinctly understood that the foregoing descriptive matter is to be interpreted merely as illustrative of the disclosure and not as a limitation.

CLAIMS

1. A prolonged release composition of Atorvastatin loaded ethosomes to enhance its bioavailability comprising of:

- atorvastatin is 10 to 20 gms;
- soya lecithin is 15 to 60ml;
- cholesterol is 15 to 45ml;
- ethanol is 5 to 15ml;
- propylene glycol is 1 to 3ml; and
- distilled water is 15 to 30ml;

wherein the drug entrapment efficiency was found to be $96.65 \pm 2.53\%$.

2. The prolonged release composition of Atorvastatin loaded ethosomes as claimed in claim 1, wherein the ethosomes were prepared by cold method followed by ultra-sonication.
3. The prolonged release composition of Atorvastatin loaded ethosomes as claimed in claim 1, wherein ethosomes were further incorporated into the gel.
4. The prolonged release composition of Atorvastatin loaded ethosomes as claimed in claim 1, wherein cumulative % of drug release in gels was found to be 98.47%.
5. A method of preparation of Atorvastatin loaded ethosomal formulation comprising the steps of:
 - i. dissolving pre-determined amounts of phospholipids, drug and other lipid material in ethanol;
 - ii. adding propylene glycol during stirring followed by heating the mixture at 30 C;
 - iii. adding 30 ml hot water into the mixture followed by stirred for 5minutes;
 - iv. using ultrasonic probe sonicator for 10 min and off time was 10 secs;
 - v. finally storing the formulation in refrigeration.
6. The method of preparation of Atorvastatin incorporated into gel comprising the steps of:
 - i. soaking 1% w/v carbopol in 5ml of water;

- ii. adding 20 ml of ethosomal suspensions containing atorvastatin 100mg;
- iii. stirring continuously at temperature of 30 C to form homogenous solution;
- iv. adding drops of triethanolamine to maintain the pH of the solution;
- v. stirring continuously to form clear transparent gel; and
- vi. finally, the ethosomal gel was mixed using a mechanical stirrer for 5 minutes.



DA van Zantwijk

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DRAWINGS

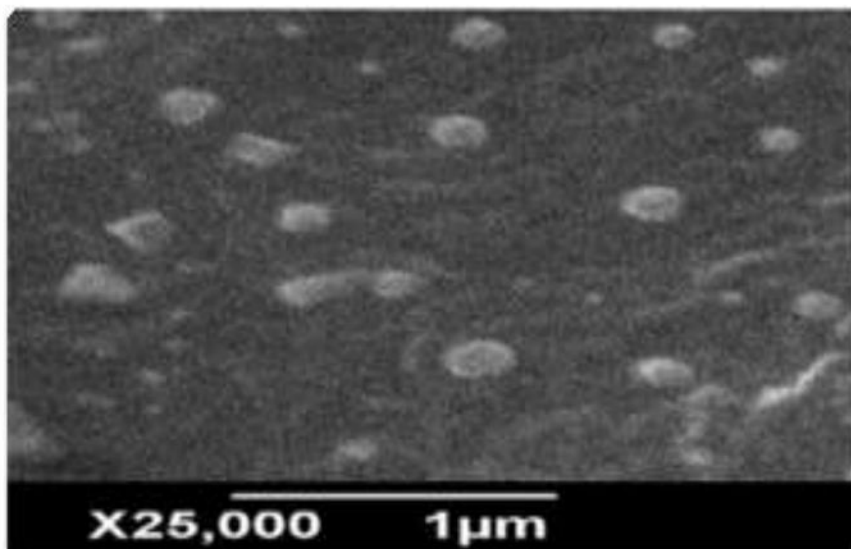


Figure 1

	Mean (mV)	Area (%)	Width (mV)
Zeta Potential (mV): -22.1	Peak 1: -18.0	63.4	6.75
Zeta Deviation (mV): 8.97	Peak 2: -30.8	36.6	4.02
Conductivity (mS/cm): 0.0995	Peak 3: 0.00	0.0	0.00

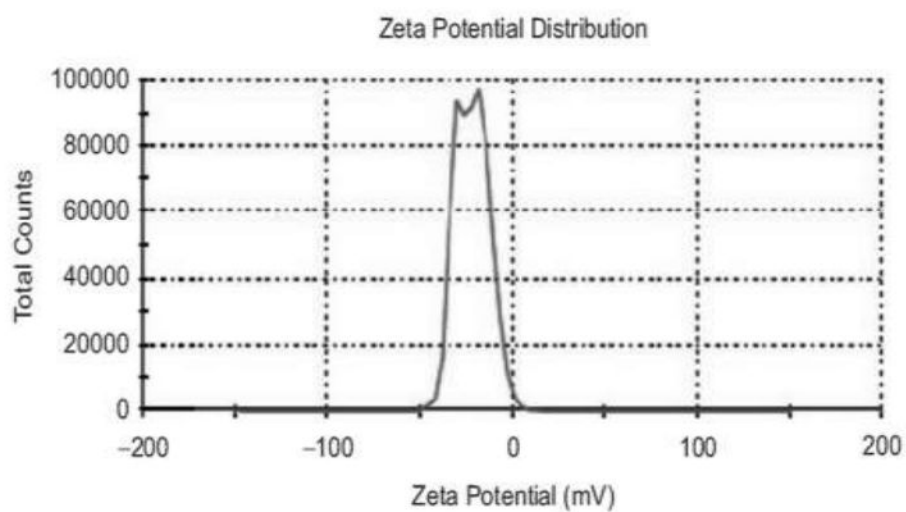


Figure 2

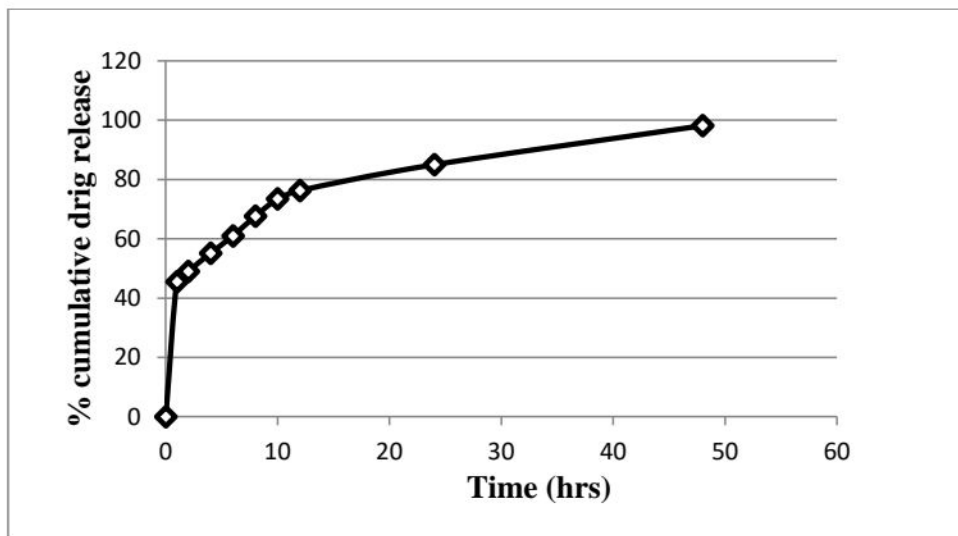


Figure 3

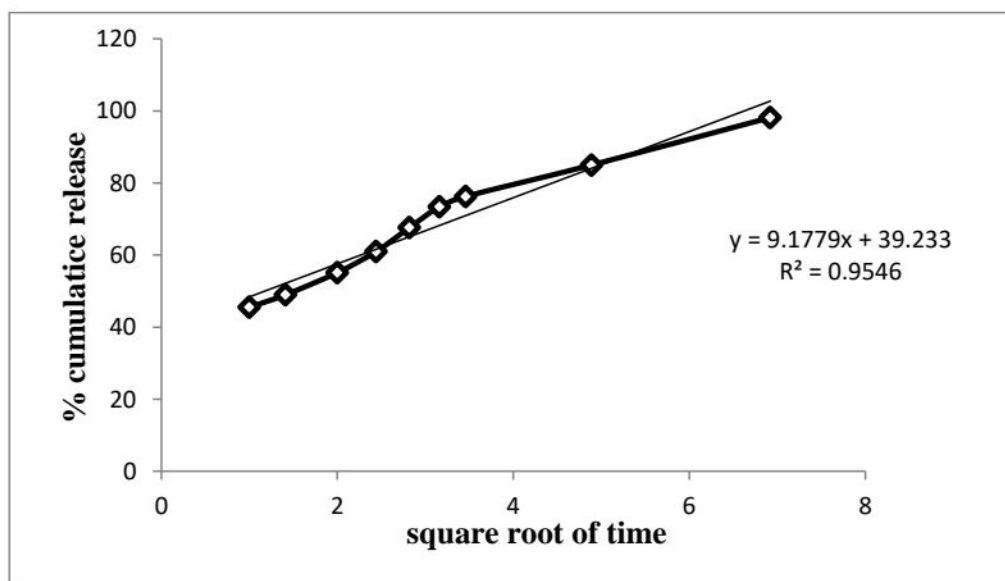


Figure 4

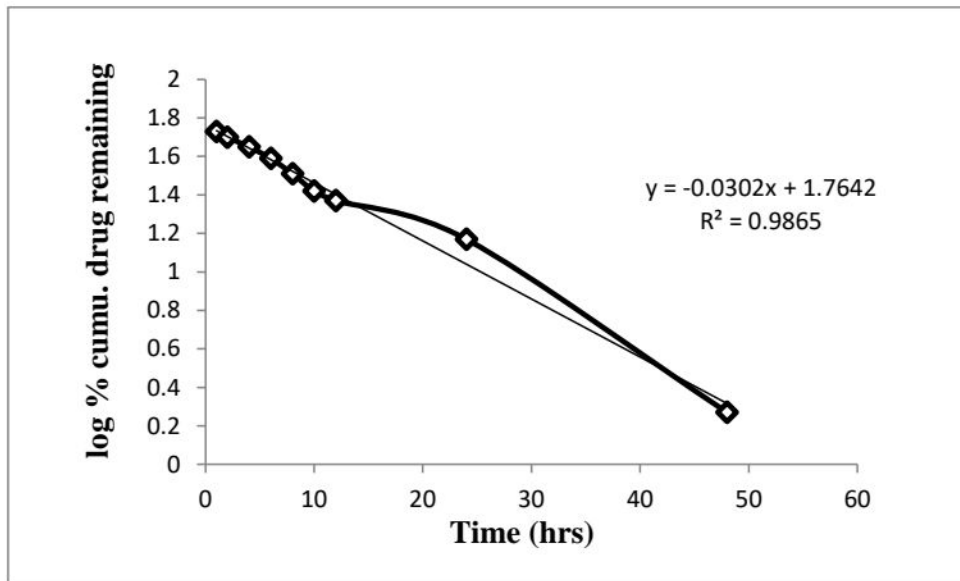


Figure 5

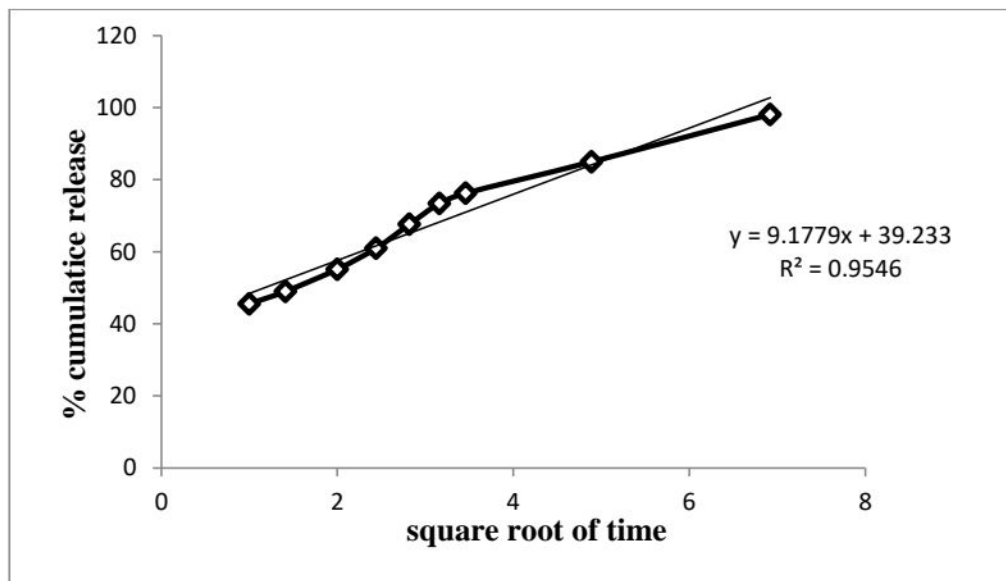


Figure 6

ABSTRACT

The present invention relates to develop a ethosomal formulation of atorvastatin to reduce the first pass metabolism and to enhance its systemic bioavailability. Atorvastatin is HMG- CoA reductase inhibitor and utilized for minimizing cholesterol level in the treatment of congestive heart failure. The prepared formulation showed enhanced drug delivery with no first pass metabolism, entrapment efficiency was found to be $87.65 \pm 2.53\%$. The gel formulation showed 15.69 gcm^2 spreadibility and 98.47% in vitro drug release within 48hrs as compared to plain drug ethosomal formulation which shows 62.37% in 48hrs. It was noticed that ethosomes would improve transdermal flux, prolong release, and provide an appealing route for long-term distribution of atorvastatin, and that the ethosomal gel delivery method is a successful design for topical drug delivery with improved bioavailability and patient enforcement.